SOVEREIGN NATIONAL INFRASTRUCTURE

RENANCE.

Nigeria's Sovereign Dual-Engine Educational Operating System

Eliminating Exam Hall Panic for Students • Ending Paper & Administrative Drain for Schools

PILLAR I: School Operations OS

- **1-Tap NERDC Auto-Install:** Primary 1 to SSS 3 curriculum mapped in 60s
- **70%+ Paper Slashing:** Single-topic ink-saving black & white lesson sheets
- **30s Attendance:** Instant classroom roll call and gate kiosk check-in
- **CA 20/20/60 & Check:** Tamper-proof grades + 6-digit PIN on /school/check

PILLAR II: Student Study OS

- **207k+ Question Arsenal:** 681 packs covering JAMB, WAEC, NECO & Uni
- **CBT Clock Coach:** Real-time Target Dwell Time pacing indicator
- **Panic Forensics:** Post-exam autopsy identifying time traps
- **Everyday Scenarios:** Explains physics via Danfos and beans

THE NATIONAL CRISIS

The Twin Bottlenecks Crippling Nigerian Education

Why Millions Fail National Exams While School Administrators Drown in Paper Chaos

Student Crisis: Clock Panic & Rote Failure

- **1.4M+ UTME Failures:** Over 75% fail cutoffs yearly; 76% struggle in WAEC science/math.
- **The Hall Clock Trap:** Students spend 4–5 minutes on single items, losing working memory and surrendering 10–15 marks to frantic blind guessing.
- **Abstract Textbook Alienation:** Dry definitions force rote memorization without physical comprehension.
- **Bandwidth Barrier:** 85% of candidates cannot afford heavy video streaming.

School Crisis: Paper Chaos & Fiscal Drain

- **Stationery Overhead:** Schools spend ₦300k+ every term on registers, lesson notebooks, and exam printing.
- **Chaotic Fee Accounting:** Disorganized records cause millions in uncollected fee arrears.
- **Report Card Forgery:** Razor-blade alterations and ₦500 cyber-cafe counterfeit prints ruin trust.
- **Teacher Burnout:** Late-night handwriting of lesson notes robs teachers of energy.

ARCHITECTURAL INNOVATION

A Sovereign Dual-Engine Operating System

Bridging Institutional Governance and Student Cognition Across 3 Native Surfaces

Pillar I: School OS

- **1-Tap NERDC Setup:** Primary 1 to SSS 3 pre-loaded in 60s.
- **Ink-Saving Notes:** Single-topic B&W sheets slash print bills by 70%+.
- **30s Attendance:** Fast mobile roll call + gate kiosk mode.
- **Timetable Matrix:** Atomic commits prevent room collisions.
- **Kobo Fee Ledger:** Integer accounting halts rounding errors.

Pillar II: Study OS

- **207k+ Question Bank:** 681 packs across JAMB, WAEC, NECO & Uni.
- **CBT Clock Coach:** Real-time pacing eliminates panic.
- **Panic Forensics:** Identifies time traps & lost marks.
- **Everyday Scenarios:** Newton's laws via Danfo buses.
- **1v1 CBT Arena:** 5-min live academic duels with WebSockets.

3 Native Surfaces

- **Web Launcher:** Next.js 15 SSG with 1,446+ edge routes.
- **Mobile Shell:** Flutter Android APK with embedded SQLite.
- **Zero Data Use:** Download once via Wi-Fi; study offline in Airplane Mode.
- **Gate Kiosk:** Plug-and-play desktop attendance terminal.
- **ADR-0003 Security:** Sealed answer keys kept server-side.

CURRICULUM VAULT

Unrivaled National Question Repository: 207,172+ Banked

681 Specialized Academic Packs Spanning 1978 to 2025 Across All Levels

JAMB UTME GATEWAY (43,553 Qs • 46 Packs)

- Official 4-subject configuration (Use of English + 3 electives)
- Realistic 120-minute countdown clock with question navigator
- Full comprehension passages, poetry analysis, and worked models

WAEC & NECO SSCE (40,077 Qs • 72 Series)

- WAEC (1988–2025): 37,696 questions with objective and theory papers
- NECO SSCE & GCE: 2,381 verified curriculum-aligned questions
- Worked theory walkthroughs illustrating official marking rubrics

POST-UTME SCREENING (Premier Universities)

- Screening papers for UNILAG, FUTA, UI, OAU, UNIBEN
- Aptitude drills, logic tests, and speed calibration for 250+ cutoffs

TERTIARY COURSE DESKS (122k+ Qs • 548 Modules)

- 100 & 200 Level foundational modules: AMS 101, COS 101, MEE 101
- Semester lecture notes, past exam pools, and CGPA

PROPRIETARY COGNITIVE ENGINE

The CBT Exam Clock Coach: Neutralizing Hall Panic

Dynamic Real-Time Pacing Guidance and Post-Exam Clock Forensics That Recover 10–15 Lost Marks

Dynamic Real-Time Pacing Engine

T_target = Total Duration / Total Questions (e.g. 60s per item)

- **[ON TRACK - GREEN]** ($t \leq 1.2 \times T_{target}$) — Steady, rhythmic, optimal velocity.
- **[LINGERING - AMBER]** ($1.2 < t \leq 2.0 \times T_{target}$) — Time buffer is eroding.
- **[TIME TRAP RISK - RED]** ($t > 2.0 \times T_{target}$) — Guess, flag, and preserve marks ahead!

Post-Exam Panic Forensics Autopsy

- **Paper Speed Curve:** Plots candidate actual pace against optimal velocity.
- **Time-Trap Isolation:** Flags questions where candidate spent >2.5× target time and still got it wrong.
- **Panic Rush Zone:** Detects end-of-test frantic guesses answered in <8s.
- **Calculated Lost Marks:** *"You lost 8 marks to clock panic, not intellectual inability."*

PEDAGOGICAL INNOVATION

Everyday Relatable Scenarios & Socratic AI

Translating Abstract Foreign Theorems into Lived Nigerian Realities and Guided Critical Thinking

Everyday Relatable Scenarios Bank

- **Newton's 1st Law (Inertia):** Standing in a speeding Danfo bus along Ikorodu Road when the driver hits the brakes — upper torso lurches forward.
- **Newton's 3rd Law (Action/Reaction):** Stepping off a wooden canoe onto muddy Lagos lagoon banks — backward push of the boat.
- **Osmosis & Membranes:** Soaking rock-hard dried honey beans in water overnight — swollen and soft by morning.
- **Opportunity Cost (Economics):** Deciding between a ₦1,500 data bundle vs a hot plate of cafeteria jollof rice.

Socratic AI Tutor (Hint-First Mode)

- **Anti-Spoon-Feeding Law:** Refuses to reveal direct option letters (A, B, C, D) or copy-paste answers, cultivating intellectual rigor.
- **Diagnostic Guided Inquiry:** Asks probing questions to isolate the core formula, variable, or rule distinction.
- **Step-by-Step Walkthroughs:** Guides candidates through equations without revealing answers.
- **Deterministic Offline Matrices:** Runs offline from local SQLite tables when internet is unreachable.

RETENTION & GAMIFICATION

1v1 Multiplayer CBT Arena & Spaced Repetition

Turning Solitary Exam Drills into an Electrifying Social Sport While Halting Cognitive Decay

1v1 Multiplayer CBT Arena (/arena)

- **5-Minute Live Duels:** 15 rapid-fire questions drawn from shared syllabi against peers nationwide.
- **Sub-100ms WebSocket Sync:** Real-time ticking scoreboards consuming negligible bandwidth.
- **Automated Sparring Bot:** Steps in within 20s if no human peer is available at odd hours.
- **Gamified XP & Streaks:** Rewards study habits and fuels national school leaderboards.

Spaced-Repetition Review Queue (/review)

- **Automated Mistake Harvesting:** Missed or skipped questions automatically route to review.
- **Ebbinghaus Curve Scheduling:** Resurfaces items at **24 Hours, 72 Hours, and 7 Days later**.
- **Targeted Remediation:** Concentrates on documented cognitive gaps instead of repeating full tests.
- **Multi-Platform Sync:** Synchronized between web portal and Android SQLite database.

TECHNICAL INFRASTRUCTURE

Zero-Data Sovereign Architecture & 5 Offline Desks

Engineered to Flourish Where Mobile Broadband Fails and Electrical Power Fluctuates Across Nigeria

The 5 Offline Desks

- **JAMB UTME Desk:** Full 46 packs + 180-Q mock generator.
- **WAEC WASSCE Desk:** 61 packs with theory walkthroughs.
- **NECO SSCE Desk:** 11 series for national secondary tests.
- **Post-UTME Desk:** Direct university screening banks.
- **School Workspace:** Teacher notes, schemes & pools.

Android SQLite Shell

- **Flutter Performance:** Fast on budget Android phones.
- **Embedded SQLite DB:** Stores 207k+ questions locally.
- **5-Minute Sync:** Connect to Wi-Fi once; study 6 months offline.
- **Airplane Mode:** CBT mocks & Clock Coach work with zero data.
- **Standalone APK:** Direct download without app store accounts.

Go Backend & Edge Web

- **Go 1.27 API:** Sub-15ms latency with async worker pools.
- **Next.js 15 SSG:** 1,446+ pre-rendered edge routes.
- **ADR-0003 Sealed Keys:** Answers kept strictly server-side.
- **SHA-256 Checksums:** Integrity verification on boot.
- **Multi-Tenant Postgres:** Complete student/school isolation.

INSTITUTIONAL EFFICIENCY

NERDC Curriculum Auto-Install & 70% Paper Slashing

Standardizing Academic Operations in 60 Seconds While Eradicating Crushing Printing Expenses

1-Tap NERDC Statutory Installation

- **Zero Manual Typing:** Provisions the official NERDC curriculum ladder in 60 seconds upon registration.
- **Primary 1 to SSS 3:** Complete basic and senior secondary subject pathways (Science, Arts, Commercial, Technical).
- **12-Week Schemes of Work:** Pre-populates term slots so teachers refine rather than starting from scratch.
- **Multi-Tier Access Control:** Proprietors manage finances; teachers access only their assigned classes.

Ink-Saving B&W Print Architecture

- **Strict Monochrome Standard:** Eliminates decorative fills and colors to optimize office laser printer life.
- **Single-Topic Print Optimization:** Prints only the specific lesson required, preventing bulk paper waste.
- **Paper Bill Reduction:** Slashes school printing overhead from ~~₦300k+~~ down to under ₦90k per academic term.
- **NERDC School Exam Pool:** 1-click terminal exam generator pulls balanced, leak-proof papers ready to print.

DAILY OPERATIONS

Attendance Roster, Gate Kiosk & Collision-Free Timetable

Streamlining Morning Roll Call, Gate Entry, and Weekly Scheduling into Synchronized Workflows

30-Second Attendance & Gate Kiosk Mode

- **30-Second Roll Call:** Mobile tap-toggle roster saves 15 minutes of instructional time each morning.
- **Gate Kiosk Mode:** Tablet/laptop at school gate logs arrivals in real time, alerting management to truancy.
- **SUBEB Compliance:** 1-click export of monthly attendance registers in CSV/JSON format for state inspections.
- **PTA Analytics:** Calculates termly student attendance percentages printed directly on verified report cards.

Collision-Free Timetable Matrix

- **Visual Weekly Grid:** Maps academic periods, assemblies, and recreation breaks across the school week.
- **Atomic Single-Transaction Commit:** Validates entire weekly schedule at once, preventing double-booked rooms.
- **Instant Staff Distribution:** Timetable immediately propagates to all teacher personal mobile dashboards.
- **Offline Reference:** Teachers view assigned schedules on their smartphones without paper printouts.

ACADEMIC INTEGRITY

Tamper-Proof Continuous Assessment & Result Checker

Eradicating Report Card Forgery and Eliminating Human Error from Terminal Grade Computation

Universal Continuous Assessment Model

Final Score (100%) = CA1 (20%) + CA2 (20%) + Terminal Exam (60%)

- **Automated Computation:** Calculates student totals, class averages, grade boundaries (A1–F9), and positions.
- **Classroom Isolation:** Teachers edit scores only for assigned cohorts, eliminating tampering.
- **Lock & Certify:** Management signs off on finalized results; records lock against retroactive edits.

Parent Result Checker (/school/check)

- **6-Digit Scratch PIN:** Unique session-scoped PIN generated for each enrolled pupil.
- **Zero-Travel Access:** Parents enter Student ID + PIN on any phone browser to unlock verified digital report cards.
- **Forgery Eradication:** Stops razor-blade grade alterations and ₦500 counterfeit cyber-cafe prints.
- **Institutional Seals:** Displays verified school emblem, remarks, and cryptographic security hashes.

FINANCIAL GOVERNANCE

Integer-Safe Kobo Accounting & Serialized ID Cards

Halting Revenue Leakage, Tracking Tuition Debtors, and Automating Student Identity Verification

Integer-Safe Kobo Accounting Ledger

- **Zero Rounding Errors:** All transactions calculated in integer kobo (100 kobo = ₦1.00).
- **Tiered Fee Structures:** Manages tuition, lab levies, uniforms, and PTA dues by class.
- **Multi-Channel Receipts:** Logs cash, bank transfers, POS, and drafts with immutable transaction hashes.
- **Real-Time Debtor Defaulter List:** Filtered roster of unpaid fees prevents exam-week evasion.

Automated Serialized ID Card Engine

- **Standardized Matriculation:** Official format: `REN-2026-NNNNNN` (e.g. `REN-2026-000142`).
- **8-Card Print Batch Sheets:** Exports printable sheets with photos, blood groups, barcodes, and seals.
- **In-House Card Production:** Schools print on cardstock and laminate in-house, saving ₦150k+ yearly.
- **Instant Revocation:** Lost or stolen cards are blacklisted without corrupting academic history.

MARKET OPPORTUNITY

Massive National Footprint Across 774 LGAs

Capturing Nigeria's Expanding EdTech Frontier with an Unfair Offline Distribution Advantage

Total Addressable Market

₦120B+

- 1.4M+ Annual JAMB UTME candidates
- 1.6M+ Annual WAEC WASSCE candidates
- 1.2M+ Annual NECO SSCE candidates
- Millions of basic education pupils in Primary 1 to JSS 3

Serviceable Market

40,000+

- Primary & secondary schools seeking digital transformation
- 250+ Tertiary Institutions nationwide
- 774 Local Government Areas: 1 sovereign rollout surface
- UBEC & SUBEB digital telemetry partnerships

Teacher Income & Scale

Two-Sided

- Earning platform for verified Nigerian teachers
- Tutors monetize revision notes and test pools
- Zero-data offline delivery reaches rural learners
- Works on budget Android phones without data bills

BUSINESS MODEL

Dual-Engine Monetization & High-Margin Unit Economics

Predictable High-Volume Consumer Subscriptions Paired with Recurring Institutional B2B SaaS

Student Consumer Economy (B2C)

- **Free Frictionless Onboarding:** 1 free full CBT test + 20 Qs/session
- **₦500 / 1 Week Pass:** Sprints for rapid pre-exam revision
- **₦1,500 / 1 Month Pass:** All 681 packs and Socratic AI tutor
- **₦2,500 / 3 Months Pass:** Termly prep for WAEC/NECO/JAMB
- **₦5,000 / 1 Year All-Access:** Full web & mobile offline access
- **REN Coin Economy:** Duels (+1 REN$), referrals (+3 REN$), redeem 250 REN$ = 1 month premium

Institutional School SaaS Tiers (B2B)

- **Basic (₦3,000/mo or ₦10,000/yr):** Up to 100 students (notes, curriculum, attendance & results)
- **Standard (₦6,000/mo or ₦25,000/yr):** Up to 300 students (exam bank, timetable, fee ledger, ID cards)
- **Premium (₦20,000/mo or ₦60,000/yr):** Unlimited students (all features + 10 free student passes)
- **Predictable Pricing:** Flat annual institutional fee; zero per-teacher micro-charges; <5% churn.

UNIT ECONOMICS: OVER 85% GROSS MARGINS — Edge-hosted SSG and local SQLite mean marginal server cost per active offline student approaches ₦0.00.

THE BUILDERS & THE ASK

Executive Leadership & ₦50M Grant Allocation

Proven Technical Leadership Shipping Sovereign Educational Infrastructure from Akure, Ondo State

The Builders: Executive Leadership

Olubobokun Femi (Resolutefemi) — CEO & Team Lead

Product Strategy, School Alliances & Regulatory Compliance

Directs school partnerships, SUBEB/NERDC stakeholder relations, and national institutional deployment across 774 LGAs.

Ayoola Damisile — CTO & Lead Systems Architect

High-Performance Go Systems & Sovereign Offline Engineering

Architect of the 681-pack offline SQLite engine, sub-15ms Go CBT scoring pipeline, and Renance AI

₦50,000,000 Capital Grant Utilization

- **₦15.0M (30%) Talent:** 10 Nigerian engineers & curriculum leads in Akure.
- **₦10.0M (20%) AI & Cloud:** Sovereign inference nodes & geo-redundant backups.
- **₦10.0M (20%) Field Rollout:** Ondo State beachhead & 50-ambassador network.
- **₦7.5M (15%) Hardware:** Low-cost test lab (Tecno/Infinix) + 5kVA solar backup.
- **₦5.0M (10%) Compliance:** CAC, trademarks, NDPR audit & NERDC certification.
- **₦2.5M (5%) Buffer:** Contingency reserve & currency inflation hedge.